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Amity University Uttar Pradesh, Greater Noida

MINOR PROJECT REPORT

(ETMN100)

WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar

Submitted by

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Declaration

I, Anubhav Anand, Enrollment No. A41105223039, hereby declare that this Minor Project report titled "WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar" is an original record of work carried out by me under the guidance of Dr. Girish Paliwal, Amity School of Engineering & Technology, Amity University Uttar Pradesh, Greater Noida, during the academic year 2026–27, and has not been submitted elsewhere for any other degree or diploma.

Faculty Guide's Certificate

This is to certify that the project report titled "WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar" submitted by Anubhav Anand (A41105223039) is a record of bona fide work carried out under my supervision in partial fulfilment of the requirements of the Minor Project (ETMN100) component of the B.Tech (Computer Science & Engineering) programme.

Dr. Girish Paliwal
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Abstract

Access to safe drinking water remains a critical public health concern across rural India, particularly in Uttar Pradesh and Bihar, where chemical and microbiological contamination of groundwater sources affects millions of households. Despite the Jal Jeevan Mission (JJM) generating extensive water quality testing data at habitation level, this information remains largely inaccessible to local district and block-level officials responsible for prioritising remediation. This project presents WaterTriage, an independently reproducible, open triage system that ingests publicly available JJM water quality data and processes it through a backend pipeline built on Python, FastAPI, SQLAlchemy, and PostgreSQL. A contamination severity scoring engine evaluates each water source against Bureau of Indian Standards (BIS) 10500 permissible limits, applying differential health-risk weights and an acute-hazard escalation rule that prevents severe contaminants such as arsenic and E. coli from being diluted by compliant aesthetic parameters. The system was validated against a canonical corpus of 1,741 live laboratory reports (966 from Uttar Pradesh, 775 from Bihar) and 100,690 historical contamination records. Results show a mean composite risk score of 18.36 for Bihar versus 12.68 for Uttar Pradesh; however, median scores (0.00 for Bihar, 4.08 for Uttar Pradesh) reveal that Bihar's elevated mean is substantially driven by a concentrated cluster of single-parameter iron panels in Madhepura district rather than uniformly elevated risk. This finding is reported transparently as a disclosed limitation of panel composition rather than a general contamination claim. The system exposes ranked, filterable intervention priority lists through a REST API and an interactive dashboard, demonstrating that independently reproducible analytics built on government open data can support ground-level water safety planning without requiring privileged access to internal government systems.

Keywords: Water Quality; Jal Jeevan Mission; Contamination Severity Scoring; BIS 10500; FastAPI; Intervention Prioritization

1. Introduction

Access to safe and adequate drinking water is a fundamental determinant of public health, yet a substantial proportion of India's rural population continues to rely on groundwater sources vulnerable to both geogenic and anthropogenic contamination. Uttar Pradesh and Bihar, situated along the Indo-Gangetic plain, are particularly affected by naturally occurring arsenic and fluoride contamination, compounded by nitrate and microbiological contamination from inadequate sanitation infrastructure. The Jal Jeevan Mission (JJM) has generated one of the largest water quality testing datasets in the country, yet this data remains consolidated within the Water Quality Management Information System (WQMIS), a reporting layer designed for national and state-level monitoring rather than direct, actionable use by district and block-level officials.

This project is motivated by the observation that publicly accessible JJM data can be independently processed to help close this accessibility gap without requiring modification to, or privileged access within, existing government systems. The objective is to design and implement WaterTriage: a reproducible system that ingests JJM water quality data for Uttar Pradesh and Bihar, scores each source against BIS 10500 limits, and produces a ranked, filterable list of intervention priorities. The two states were selected for both practical feasibility and their genuinely distinct contamination geographies, offering a meaningful basis for comparative analysis.

2. Related Work

2.1 Government Water Quality Monitoring Infrastructure. WQMIS, established under JJM by the Department of Drinking Water and Sanitation, consolidates water quality test results at state and district laboratories. Published accounts describe this infrastructure as operating primarily for programme-level monitoring rather than generating ranked, ground-level intervention priorities for block or panchayat-level functionaries.

2.2 Groundwater Contamination in the Middle Gangetic Plain. Hydrogeochemical research has documented severe arsenic and fluoride contamination across this region, with arsenic concentrations reported up to several thousand µg/L in parts of the middle Gangetic floodplain and fluoride concentrations up to 12 mg/L in affected Bihar districts, substantially exceeding BIS permissible limits.

2.3 Water Quality Index and Contamination Scoring Approaches. Conventional Water Quality Index (WQI) methodologies are known to suffer from an "eclipsing" effect, whereby a single highly toxic parameter exceeding safe limits can be masked by a composite score dominated by compliant parameters. Reviews of WQI methodology find no consensus on universal parameter weighting, concluding that weighting should be adapted to local health priorities rather than treated as fixed across contexts.

2.4 Positioning of This Project. WaterTriage applies a BIS 10500-grounded, severity-weighted scoring approach explicitly designed to avoid the eclipsing flaw, to JJM monitoring data for Uttar

Pradesh and Bihar, with the goal of producing actionable output that existing government reporting systems do not directly provide to local stakeholders.

3. Methodology

3.1 System Architecture. WaterTriage is implemented as a four-stage pipeline: acquisition, storage, enrichment, and serving. Data is acquired from public WQMIS reports and cached locally, persisted in a normalized PostgreSQL schema, enriched with per-record severity scores, and served through a REST API and browser dashboard.

3.2 Database Schema. The system persists data across eight tables: a geographic hierarchy (states, districts, villages), bis_parameters (BIS limits and health weights), water_samples and readings (separating a lab report from its individual parameter measurements), risk_scores, historical_contamination (registry events from 2009–2012), and interventions.

3.3 Contamination Severity Scoring. Each parameter reading is converted to a severity value between 0 and 1 by linear interpolation between the BIS acceptable and permissible limits. During development, iron and nitrate were initially found to have no defined permissible ceiling in the scoring configuration, which caused any exceedance of the acceptable limit to saturate directly to maximum severity (a binary rather than graded response). This was identified as a methodological defect and corrected by assigning iron a BIS-aligned permissible limit of 1.0 mg/L, restoring graded linear interpolation consistent with every other parameter. The composite risk score (CRS) for a sample is the weighted mean of severity across tested parameters, scaled to 0–100, using health-risk weights (arsenic = 1.0, E. coli = 0.9, fluoride = 0.8, nitrate = 0.7, iron = 0.4, turbidity/TDS = 0.3, pH = 0.2). A band-escalation rule additionally forces a minimum "High" classification for any parameter reaching full severity, and a "Critical" classification for acute-hazard parameters (E. coli, arsenic ≥ 0.05 mg/L) irrespective of the numeric composite score. Based on CRS, each sample is classified as Critical (≥75), High (50–74), Medium (25–49), or Low/Safe (<25).

3.4 Recurrence Detection. A recurrence classification module was implemented to blend historical registry presence with current exceedance status, producing verdicts of chronic-hotspot, recurring, watchlist, or no-history for a given village-parameter combination. This logic was validated through unit testing in isolation; however, live linkage between the production water_samples table and the historical_contamination registry could not be established for this reporting period because the two data sources use disjoint village identifier schemes. This limitation is disclosed in full in Section 6.

3.5 Data Acquisition. The primary acquisition channel is a crawler against WQMIS's public, unauthenticated report-printing endpoint, operating with caching and resume checkpoints. A historical Quality-Affected Habitations registry (2009–2012, 100,690 events) was loaded as a supplementary source.

3.6 Implementation and Validation. The backend is implemented in Python using FastAPI and SQLAlchemy against PostgreSQL, containerized via Docker Compose. A dedicated /api/meta endpoint serves corpus-level invariants as a single source of truth for dashboard consumers, shown in Table 3. The implementation was validated using twenty-eight automated test cases (one skipped), covering scoring correctness, API contract behaviour, WQMIS parser edge cases, and database integration.

Invariant	**Description**
total_samples == 1,741	Corpus size matches the frozen canonical count
band_counts.sum() == total_samples	Severity band distribution is exhaustive
critical_count == 89	Critical-band count matches the corrected scoring run
readings_total == 6,792	Reading count matches the corpus

4. Results

4.1 Data Volume. The canonical, frozen corpus used for all results in this section comprises 1,741 scored live laboratory reports (Uttar Pradesh: 966; Bihar: 775), 6,792 discrete parameter readings, and 100,690 historical contamination registry events. This figure supersedes several intermediate sample counts (755, 1,245, and 1,855) reported in earlier weekly progress reports during active development; those counts reflected partial crawls and an interim SQLite-to-PostgreSQL migration and are not used in this final report.

4.2 Severity Band Distribution. Table 1 shows the distribution of the 1,741 scored samples across severity bands under the corrected scoring methodology (Section 3.3).

Severity Band	Count	Share
Critical (CRS ≥ 75)	89	5.1%
High (50 ≤ CRS < 75)	392	22.5%
Medium (25 ≤ CRS < 50)	28	1.6%
Low / Safe (CRS < 25)	1,232	70.8%
Total	1,741	100%

4.3 Comparative State Analysis: Mean, Median, and a Panel-Size Artifact. Table 2 compares risk statistics between the two states.

Statistic	Uttar Pradesh (n=966)	Bihar (n=775)
Mean composite risk score (CRS)	12.68	18.36
Median composite risk score (CRS)	4.08	0.00
Critical band samples (CRS ≥ 75)	14	75
High band samples (50 ≤ CRS < 75)	234	158

Bihar's mean composite risk score (18.36) exceeds Uttar Pradesh's (12.68); however, Bihar's median score is 0.00, compared to 4.08 for Uttar Pradesh, indicating that more than half of Bihar's scored samples fall in the Low/Safe band. This divergence between mean and median is explained by a specific data artifact: 70 of the 89 Critical-band samples (78.7%) originate from Madhepura district alone, where all 70 samples were tested under a single-parameter panel for iron only. Under the corrected scoring rule, any iron reading at or above the permissible limit saturates to maximum severity, and because no other parameters were tested for these samples, the composite score for each equals 100 by construction. This is disclosed explicitly as a panel-composition limitation rather than evidence of uniformly severe contamination across Bihar, and is surfaced to dashboard consumers via the /api/meta endpoint's documented limitations.

4.4 Escalation Rule Validation. During testing, synthetic and live cases confirmed that the band-escalation rule (Section 3.3) correctly reclassifies samples with acute hazards (E. coli presence or arsenic ≥ 0.05 mg/L) to Critical band even where the arithmetic composite score alone would classify the sample as Low or Medium, confirming the rule functions as intended to prevent the eclipsing failure mode discussed in Section 2.3.

4.5 Recurrence Module Status. The historical registry confirms Khokha village (Purnia district, Bihar) as the single most frequently recorded location for iron-related events in the 2009–2012 registry,

with 166 recorded events — the highest of any village in the historical dataset. However, because the production water_samples table and the historical_contamination registry use disjoint village identifier schemes, this historical finding could not be linked to a corresponding live risk score for the same village during this reporting period, and an earlier characterisation of Khokha as a "confirmed chronic contamination hotspot" based on live system output has accordingly been withdrawn. The historical figure is reported here as a registry-level finding only, and the identifier-mapping gap is recorded as an open limitation (Section 6).

4.6 System Validation. All twenty-eight automated tests pass (one skipped), and the four corpus-level invariants exposed via /api/meta (Table 3) were confirmed to hold against the production database at the time of this report.

5. Discussion

The results in Section 4.3 illustrate why reporting both mean and median statistics, together with panel-composition context, is necessary for an honest comparative analysis: a mean-only reading would have suggested Bihar carries substantially higher contamination risk than Uttar Pradesh across the board, when the more complete picture is that a small, structurally distinct cluster of single-parameter samples accounts for the great majority of that gap. This is treated in this report as a methodological finding about the dataset's composition, not a claim about underlying contamination incidence rates between the two states, which this sample is not equipped to estimate without correcting for panel-size variation.

The correction to the iron/nitrate scoring logic (Section 3.3) was itself a significant finding: the original binary-saturation behaviour for parameters lacking an explicit permissible limit caused disproportionate Critical-band inflation, which was only identified through systematic re-verification of the scoring engine against its own BIS 10500 justification. This underscores the value of treating a scoring model's assumptions as testable claims rather than fixed infrastructure.

The withdrawal of the Khokha chronic-hotspot claim (Section 4.5) reflects the same discipline: rather than retaining a compelling but unverifiable narrative example, the identifier-mismatch limitation is disclosed directly, and the historical finding is reported at the scope it can actually support.

6. Limitations

This work is bounded by several limitations that should inform how its findings are used. First, the 1,741 live laboratory reports represent a sample drawn from targeted WQMIS identifier ranges rather than a spatially uniform census of either state; sample density by district reflects crawler coverage, not necessarily underlying population-level contamination burden. Second, as discussed in Section 4.3, a subset of Critical-band samples originate from single-parameter test panels (iron only), which mechanically produces maximum severity scores independent of a source's overall water quality and should not be read as representative of multi-parameter contamination severity. Third, live linkage

between the historical contamination registry and current water_samples records could not be established due to disjoint village identifier schemes between the two sources, limiting recurrence/trend classification to registry-level findings rather than sample-linked verdicts for this reporting period. Fourth, a temporal discontinuity exists between the historical registry (2009–2012) and current WQMIS testing cycles, meaning historical findings indicate long-term hydrogeological susceptibility rather than real-time contamination status.

7. Conclusion

This project set out to design and implement WaterTriage, a system that ingests publicly available JJM water quality data for Uttar Pradesh and Bihar and produces ranked, actionable intervention priorities that existing government reporting systems do not directly provide to local officials. Table 4 maps the objectives declared in the project synopsis against delivered outcomes.

Synopsis Objective	**Delivered Outcome**
Acquisition and normalization of JJM open water-quality data for UP and Bihar	Achieved — WQMIS public report crawler delivering 1,741 scored live laboratory reports (UP 966 / Bihar 775), supplemented by a 100,690-event historical registry (2009–2012)
Relational database schema for source-level records	Achieved — eight-table normalized PostgreSQL schema with canonical state naming and referential integrity
Contamination severity scoring model per BIS IS 10500	Achieved — weighted per-parameter severity with linear interpolation between acceptable and permissible limits (including a corrected iron permissible limit of 1.0 mg/L) and an acute-hazard escalation rule
Recurrence/trend detection module	Partially achieved — scoring and classification logic implemented and unit-tested in isolation; live linkage to the historical registry is blocked by disjoint village identifiers between the two data sources (Section 6, Limitations)
FastAPI backend exposing ranked intervention lists	Achieved — REST endpoints including ranked priority queue, state comparison, and a dedicated /api/meta endpoint serving verified corpus invariants
Dashboard for district-wise and state-wise comparison	Achieved — interactive dashboard with priority queue, overview KPIs, and methodology/limitations panel sourced from /api/meta
Containerized deployment	Partially achieved — PostgreSQL and backend services containerized via Docker Compose

The system was implemented, validated against twenty-eight automated test cases, and evaluated against a reconciled, canonical database of 1,741 real laboratory reports and 100,690 historical contamination records. Over the course of implementation, two significant methodological issues were identified and corrected: a scoring-logic defect causing disproportionate Critical-band inflation for certain parameters, and an unverifiable recurrence claim that was withdrawn once a data-linkage limitation was identified, rather than retained uncorrected. The system's central contribution is an open, independently reproducible triage layer that renders JJM water-quality data actionable for district and block-level officials, demonstrating, within the bounds of an 82-day Minor Project, that rigorous data-engineering practice — including the discipline of identifying and disclosing one's own errors — can be meaningfully applied to a socially consequential public-health problem. Future work includes resolving

the village-identifier mismatch to enable true sample-linked recurrence detection, expanding live crawler coverage, and completing production hardening of the deployment.

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